

CRPS-Optimal Binning for Univariate Conformal Regression

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Abstract

We propose a non-parametric, model-free method for conditional distribution estimation based on partitioning covariate-sorted observations into contiguous bins and using the within-bin empirical CDF as the predictive distribution. Bin boundaries are chosen to minimise the total leave-one-out Continuous Ranked Probability Score (LOO-CRPS), which admits a closed-form cost function with $O(n^2 \log n)$ precomputation and $O(n^2)$ storage; the globally optimal K -partition is recovered by a dynamic programme in $O(n^2 K)$ time. We select K by K -fold cross-validation of test CRPS, which yields a U-shaped criterion with a well-defined minimum. Having selected K^* and fitted the full-data partition, we form a conformal prediction set based on CRPS as the nonconformity score, which carries a finite-sample marginal coverage guarantee at any prescribed level ε . The conformal prediction is transductive and data-efficient, as all observations are used for both partitioning and p-value calculation, with no need to reserve a hold-out set. On real-world small-data benchmarks against split-conformal competitors (Gaussian split conformal, CQR, CQR-QRF, CHR, and conformalized isotonic distributional regression), the full- n method produces narrower prediction intervals while maintaining near-nominal coverage; the advantage is not as clear-cut however in a matched-sample comparison restricted to the same training half as the competitors. Similarly, a comparison against QOOB, another transductive competitor, provides mixed results.

Keywords: conformal prediction, conformal regression, distribution-free, non-parametric regression, optimal binning

1. Introduction

A fundamental problem in supervised learning is to estimate not just the conditional mean $\mathbb{E}[Y \mid X = x]$ but the full conditional distribution $P(Y \mid X = x)$. A distributional forecast quantifies uncertainty and therefore enables principled decision-making under risk, hypothesis testing at a test point, and the construction of valid prediction sets.

The simplest non-parametric approach is to collect nearby training observations and use their Empirical Cumulative Distribution Function (ECDF) as a stand-in for $P(Y \mid X = x)$. Other approaches exist, e.g. Kernel density methods, but they have high sensitivity to bandwidth choice or significant computational requirements.

When the covariate is one-dimensional and the data are sorted by x , a natural alternative is to partition the sorted sequence into contiguous bins and predict with the within-bin ECDF.

The central question then is how to place bin boundaries optimally. Informal choices, such as equal-width or equal-count bins, are oblivious to the heteroscedasticity of the response and to the local density of the covariate. We argue that bin boundaries should minimise a proper scoring rule evaluated on the training data, so that the binning criterion and the predictive goal are aligned. Among proper scoring rules, the Continuous Ranked Probability Score (CRPS) is particularly well suited to this purpose: it targets the full distribution (not a single functional), it admits a closed-form leave-one-out formula that depends only on pairwise absolute differences, and it is the same score used in the conformal prediction step, ensuring coherence between bin selection and prediction-set construction.

Contributions.

1. **Closed-form LOO-CRPS cost.** We derive the total leave-one-out CRPS of a bin of size m as $\text{cost}(S) = mW/(m-1)^2$, where $W = \sum_{\ell < r} |y_\ell - y_r|$ is the pairwise dispersion (Proposition 1). This scalar is updated in $O(\log n)$ per extension step, giving $O(n^2 \log n)$ precomputation and $O(n^2)$ storage for all subintervals.
2. **Exact optimal partitioning via dynamic programming.** The additive cost structure satisfies the optimal substructure property, so a DP recovers the globally optimal K -partition in $O(n^2 K)$ time (Section 5). Unlike greedy binary segmentation (Auger and Lawrence, 1989), which fixes cut points sequentially and can miss the global optimum, the DP evaluates all continuations simultaneously and is exact — and vastly more efficient than exhaustive search over all $\binom{n-1}{K-1}$ candidate K -partitions.
3. **Conformal prediction sets.** We construct CRPS-based conformal prediction sets with finite-sample marginal coverage guarantees. In the split-conformal case, convexity of the score guarantees that prediction sets are always connected intervals; in our transductive setting, single-interval structure is observed empirically in all experiments. The regression form of the Venn Predictor (a constant-width family of augmented ECDFs) is formalised as a complementary output in Appendix D.

A notable feature of the method is that it is *transductive*: all n observations are used for both partitioning and conformal calibration, with no data held out. Split-conformal competitors must reserve a calibration set, effectively halving the sample available for model fitting; this data-efficiency advantage is particularly pronounced in small-sample settings, though not unique to this construction, as most other conformal procedures can equally be cast in a full/ transductive (or jackknife+/CV+) form. (Section 9 compares against QOOB (Gupta et al., 2022), another transductive method.) The $O(n^2 K)$ dynamic programme is therefore most practical, and the transductive efficiency advantage most valuable, in the small-to-moderate- n regime that motivates this work (both real-data experiments in Section 9 have $n < 300$); for much larger datasets a split-conformal variant, or a coarser approximate DP, would be preferable. The method is currently restricted to a one-dimensional covariate; Section 10 discusses two natural extensions to $d > 1$ (a learned one-dimensional projection, and CART-style greedy splitting) along with their tractability trade-offs.

2. Related Work

Conformal prediction for regression. Conformal prediction (Vovk et al., 2022) wraps any nonconformity score in a distribution-free prediction set with finite-sample marginal coverage; Lei et al. (2018) provide a comprehensive treatment for the regression setting, covering a range of nonconformity scores and establishing finite-sample guarantees. The inductive (split-conformal) variant (Papadopoulos et al., 2002) divides training data into a fitting half and a calibration half; the quantile of calibration nonconformity scores determines the prediction set for a new point. Conformalized Quantile Regression (CQR) (Romano et al., 2019) improves efficiency in heteroscedastic settings by using conditional quantile estimates as the base predictor and measuring residuals relative to the estimated interval. Mondrian conformal prediction (Vovk et al., 2022) stratifies the calibration set into categories (taxonomies) so that coverage holds conditionally within each stratum; our method is a Mondrian predictor with data-adaptive bins as strata and CRPS as the nonconformity score. Sesia and Romano (2021) (CHR) build a conditional histogram of the response from a black-box quantile regressor, then take the shortest contiguous run of histogram bins reaching the target coverage, conformalized on a held-out calibration set (Appendix H details our reimplementation); our method instead partitions on the covariate x directly, selecting bin boundaries via a closed-form CRPS-optimal criterion, and calibrates transductively rather than via a data split (Section 9 compares both empirically). Bao et al. (2025) survey and empirically compare eight univariate conformal regression methods, providing broader benchmarking context for the split-conformal competitors evaluated here. Luo and Zhou (2025) construct efficient conformal intervals by thresholding estimated conditional interquantile densities, targeting the Neyman–Pearson-optimal set size via a different route than CRPS-based binning. Gao et al. (2025) give a distribution-free dynamic-programming algorithm for near-volume-optimal prediction sets restricted to unions of k intervals in the response space; our dynamic program instead partitions the covariate space into contiguous bins and is tied specifically to the CRPS-optimal criterion rather than general volume minimisation.

Venn predictors. Venn predictors output a set of probability distributions, one for each candidate label of the test point. This work draws much of its inspiration from the Venn-ABERS predictors for binary classification proposed by Vovk and Petej (2014): isotonic regression on an underlying scoring function implicitly identifies an optimal partition of the score range into reference classes, each with finite-sample calibration guarantees. Our method mirrors this structure on the covariate axis: the DP selects CRPS-optimal bins, defining reference classes whose within-bin ECDF best describes the local conditional distribution.

Conformal predictive systems and distributional conformal prediction. Allen et al. (2025) establish a unifying framework: any distributional regression procedure that is in-sample calibrated, when conformalized, yields a conformal predictive system with out-of-sample calibration guarantees under exchangeability. Their framework covers conformal binning, that is, partitioning calibration data into covariate-similar groups and predicting with within-group ECDFs, and conformal isotonic distributional regression (IDR) as canonical instances, but leaves the bin-selection criterion unspecified. The present paper fills this

gap: we identify CRPS-optimal contiguous bins via dynamic programming and provide a closed-form cost formula, a CV criterion for K , and structural results on the resulting prediction set. Since the within-bin ECDF is in-sample auto-calibrated by construction, our conformal binning step is an instance of their framework and inherits the corresponding calibration guarantee. Chernozhukov et al. (2021) construct (approximately) conditionally valid prediction intervals by permuting PIT residuals; their approach targets conditional validity via a rank-based argument rather than CRPS-optimal predictive distributions. Randahl et al. (2026) enforce coverage within pre-specified *outcome* bins to promote coverage equity across label subgroups; their partition is on the response y rather than on the covariate x , and the objective is coverage equity rather than predictive sharpness.

Conditional distribution estimation. Model-free competitors include Quantile Regression Forests (Meinshausen, 2006) and NGBoost (Duan et al., 2020); unlike these, our method requires only observations sorted by covariate and makes no distributional assumption.

Optimal partitioning. The DP recurrence (Section 5) is structurally identical to exact segment neighbourhood algorithms (Auger and Lawrence, 1989; Killick et al., 2012). The key distinction is objective: classical methods seek distributional homogeneity, whereas we minimise predictive LOO-CRPS.

3. Setup

Let $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R}^2$ be a training sample. Sort observations by covariate value so that $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$, and write y_i for the response paired with $x_{(i)}$.

A K -partition of the sorted observations is a sequence of indices $0 = b_0 < b_1 < \dots < b_K = n$ defining K contiguous bins $B_k = \{b_{k-1} + 1, \dots, b_k\}$ for $k = 1, \dots, K$. Within bin B_k , the predictive distribution for a new $x \in [x_{(b_{k-1}+1)}, x_{(b_k)}]$ is taken to be the empirical CDF of $\{y_i : i \in B_k\}$.

4. LOO-CRPS Cost of a Bin

The CRPS: definition and geometric interpretation

For a predictive CDF F and a scalar outcome $y \in \mathbb{R}$, the *Continuous Ranked Probability Score* (Gneiting and Raftery, 2007) is the integrated squared difference between F and the CDF of a point mass at y :

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(t) - \mathbf{1}[t \geq y])^2 dt. \quad (1)$$

Here $\mathbf{1}[t \geq y]$ is the right-continuous CDF of a Dirac mass at y ; some references write $\mathbf{1}[t < y]$ (the left-continuous version), which agrees everywhere except at the single point $t = y$ and leaves the integral unchanged. Geometrically, the integrand is the squared vertical gap between F and the step at y (Figure 1). CRPS is zero if and only if F is a point mass at y ; diffuse or mis-centred forecasts accumulate a larger integrated gap and hence a larger

score. An equivalent energy-score form (Gneiting and Raftery, 2007, Eq. (21)), convenient for computation, is

$$\text{CRPS}(F, y) = \mathbb{E}_F |X - y| - \frac{1}{2} \mathbb{E}_F |X - X'|, \quad (2)$$

where X, X' are independent draws from F . The first term penalises mis-location; the second subtracts a self-dispersion penalty that rewards sharpness, preventing the score from being minimised by a diffuse prior. CRPS is a *strictly proper* scoring rule for the class of all distributions with finite first moment (Gneiting and Raftery, 2007): its expected value under P is uniquely minimised by the forecast $\hat{F} = P$.

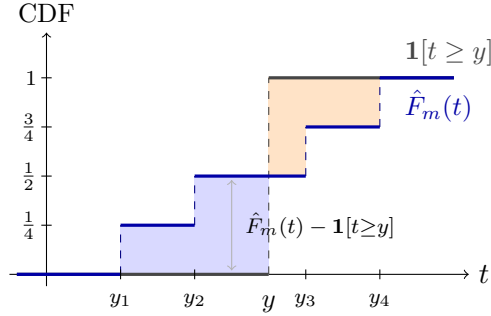


Figure 1: Geometric interpretation of $\text{CRPS}(\hat{F}_m, y)$ as the integral of the squared vertical gap between the predictive CDF $\hat{F}_m$ (blue step function, $m = 4$ atoms) and the step $\mathbf{1}[t \geq y]$ (grey) at the observed outcome y . Blue shading marks intervals where $\hat{F}_m(t) > \mathbf{1}[t \geq y]$ (too little forecast mass above t); orange marks intervals where $\hat{F}_m(t) < \mathbf{1}[t \geq y]$ (too little mass below t). $\text{CRPS} = \int (\hat{F}_m(t) - \mathbf{1}[t \geq y])^2 dt$ is large when the forecast is mis-centred or over-dispersed relative to y .

Empirical CDF and the LOO cost

For a predictive CDF $\hat{F}$ consisting of m equally weighted atoms and outcome y , applying (2) gives

$$\text{CRPS}(\hat{F}, y) = \frac{1}{m} \sum_{i=1}^m |y_i - y| - \frac{1}{2m^2} \sum_{i=1}^m \sum_{j=1}^m |y_i - y_j|.$$

Let S be a bin of size m with response values $y_1, \dots, y_m$. Write

$$d_k = \sum_{\ell \neq k} |y_\ell - y_k|, \quad D = \sum_{\ell \neq r} |y_\ell - y_r| = 2 \sum_{\ell < r} |y_\ell - y_r|.$$

The leave-one-out predictive distribution for observation k is the ECDF of $\{y_\ell : \ell \in S, \ell \neq k\}$. Applying the CRPS formula with $m - 1$ atoms:

$$\text{CRPS}(\hat{F}_{S \setminus \{k\}}, y_k) = \frac{d_k}{m-1} - \frac{D - 2d_k}{2(m-1)^2}.$$

Proposition 1 *The total leave-one-out CRPS of bin S is*

$$\text{cost}(S) = \sum_{k \in S} \text{CRPS}(\hat{F}_{S \setminus \{k\}}, y_k) = \frac{m}{2(m-1)^2} D = \frac{m}{(m-1)^2} \sum_{\ell < r, \ell, r \in S} |y_\ell - y_r|.$$

Proof See Appendix A. ■

For a bin spanning sorted indices i through j (with $m = j - i + 1$), write

$$W(i, j) = \sum_{\substack{\ell < r \\ \ell, r \in \{i, \dots, j\}}} |y_\ell - y_r|, \quad c(i, j) = \frac{j - i + 1}{(j - i)^2} W(i, j).$$

5. Dynamic Programme

It turns out that it is possible to solve the optimal partitioning problem using dynamic programming. The DP is efficient compared to the combinatorial explosion of exhaustive search, which would enumerate all $\binom{n-1}{K-1}$ candidate K -partitions, and is exact, unlike, say, greedy binary segmentation, which fixes cut points sequentially and can miss the global optimum. The problem of minimizing the total LOO-CRPS of a K -partition can be solved by a DP because the cost function can be expressed in a recurrence that exhibits the optimal substructure property, as we will show below.

Define $\text{dp}[k][j]$ as the minimum total LOO-CRPS achievable by partitioning observations $1, \dots, j$ into exactly k contiguous bins.

Minimum bin size. The LOO cost $c(i, j)$ requires at least two observations: for a singleton bin the leave-one-out distribution is empty, so the LOO-CRPS is undefined. We set $c(i, i) = +\infty$ by convention, which causes the DP to exclude singleton bins automatically. The index ranges below implicitly assume $j - i \geq 1$ (i.e. $m \geq 2$).

Base case.

$$\text{dp}[1][j] = c(1, j), \quad j = 2, \dots, n.$$

Recurrence. If $\text{dp}[k-1][i]$ gives the cost of the optimal partition of the first i observations into $k-1$ bins, then to partition $1, \dots, j$ into k bins optimally it suffices to try every candidate last-bin start $i+1$: the last bin covers observations $i+1, \dots, j$ with cost $c(i+1, j)$, and the preceding i observations are already optimally split at cost $\text{dp}[k-1][i]$. Taking the minimum over i gives the recurrence ($k \geq 2, j \geq k$):

$$\text{dp}[k][j] = \min_{k-1 \leq i < j} \left\{ \text{dp}[k-1][i] + c(i+1, j) \right\}.$$

Solution. The optimal K -partition cost is $\text{dp}[K][n]$. Bin boundaries are recovered by backtracking the argmin at each step.

The complete pseudocode is given as Algorithm 1 in Appendix B; a Python implementation is also available in the `crpsconfreg` package.

6. Precomputation and Complexity

Precomputing $W(i, j)$. Fix left endpoint i and scan $j = i, i+1, \dots, n$. Let $m = j - i + 1$ denote the current number of elements in the bin $[i, j]$. Adding y_{j+1} to the set $\{y_i, \dots, y_j\}$ increases the pairwise dispersion W by $\Delta W = \sum_{\ell=i}^j |y_\ell - y_{j+1}|$. To evaluate this sum in $O(\log n)$ without iterating over all m elements, split by sign. Let r denote the number

of existing values $\leq y_{j+1}$ (the *rank* of y_{j+1} in the current set), $S_{\leq} = \sum_{\ell: y_{\ell} \leq y_{j+1}} y_{\ell}$, and $S_{>} = \sum_{\ell: y_{\ell} > y_{j+1}} y_{\ell}$. Then

$$\begin{aligned} \Delta W &= \sum_{\ell: y_{\ell} \leq y_{j+1}} (y_{j+1} - y_{\ell}) + \sum_{\ell: y_{\ell} > y_{j+1}} (y_{\ell} - y_{j+1}) \\ &= y_{j+1} \cdot r - S_{\leq} + S_{>} - y_{j+1} \cdot (m - r). \end{aligned}$$

The quantities r and $S_{\leq}$ are *prefix queries*: a count and a sum over all stored values up to a query point. $S_{>}$ follows as the running total minus $S_{\leq}$. Prefix queries can be supported efficiently using Fenwick trees (Binary Indexed Trees) (Fenwick, 1994), for which searches and insertions can be performed in $O(\log n)$ time. We use two Fenwick trees (one for counts, one for values) indexed by value rank. Since there are $O(n)$ left endpoints and $O(n)$ extensions for each, the total precomputation is $O(n^2 \log n)$ with $O(n^2)$ storage. Data-structure trade-offs are discussed in Appendix B.

DP. The recurrence evaluates in $O(n^2 K)$ time. There are K layers and $O(n)$ entries per layer; filling each entry $\text{dp}[k][j]$ requires scanning $O(n)$ candidate split points i , with each lookup of $c(i+1, j)$ taking $O(1)$ from the precomputed table.

Complexity tightness. In some cases, the efficiency of the DP can be further improved by exploiting the structure of the cost function. A standard result of algorithm theory often referred to as the Knuth–Yao theorem (Knuth, 1971; Yao, 1980) indicates that if the cost function satisfies a quadrangle inequality, then the DP can be further sped up. Unfortunately, the LOO-CRPS cost violates the quadrangle inequality (explicit counterexample in Appendix C), so, at least under this respect, $O(n^2 K)$ is tight: no Knuth–Yao speedup applies.

7. Selecting K

The DP finds the optimal partition, i.e. the one that minimises total LOO-CRPS, for a *fixed* K . A natural approach to selecting K is to treat the total LOO-CRPS $\text{dp}[K][n]$ as a function of K and pick its minimum¹. As it is customary, we perform cross-validation to strike a balance between underfitting and overfitting. The sorted observations are assigned to folds by interleaving: observation i (in the x -sorted order) goes to fold $i \bmod F$, so that every fold inherits the x -sorted structure (note that we use F for the number of folds to avoid confusion with the number of bins K). For each fold $f \in \{0, \dots, F-1\}$, let $\mathcal{T}_f$ and $\mathcal{V}_f$ denote the training and test subsets. For each candidate K , find the optimal K -partition $\hat{\mathcal{P}}_K^{(f)}$ on $\mathcal{T}_f$ using the DP, then evaluate the test CRPS on $\mathcal{V}_f$:

$$\text{TestCRPS}_f(K) = \frac{1}{|\mathcal{V}_f|} \sum_{(x_i, y_i) \in \mathcal{V}_f} \text{CRPS}\left(\hat{F}_{b(x_i)}^{(f)}, y_i\right),$$

1. As an alternative, one could place a Dirichlet process prior over the partition; the number of clusters is then determined by the concentration parameter rather than cross-validation. While principled, this approach does not in general minimise CRPS, and the choice of concentration parameter introduces a separate hyperparameter. We use CV as a simpler, assumption-free alternative.

where $b(x_i)$ is the bin in $\hat{\mathcal{P}}_K^{(f)}$ that contains x_i and $\hat{F}_{b(x_i)}^{(f)}$ is the ECDF of the training y -values in that bin. The final criterion averages across folds:

$$\overline{\text{TestCRPS}}(K) = \frac{1}{F} \sum_{f=0}^{F-1} \text{TestCRPS}_f(K), \quad K^* = \arg \min_{K=1, \dots, K_{\max}} \overline{\text{TestCRPS}}(K).$$

We use $F = 5$ folds and set $K_{\max} = \lfloor n/10 \rfloor$ throughout, ensuring the minimum expected bin size on each training fold is at least 8; values larger than $\lfloor n/10 \rfloor$ typically produce nearly empty bins and are penalised heavily by test CRPS regardless.

Running example. To illustrate the methods of this and subsequent sections, we use a heteroscedastic synthetic dataset with $n = 1000$ observations: $X_i \sim \text{Uniform}(0, 3)$, $Y_i \mid X_i = x \sim \mathcal{N}(3x, (1+x)^2)$. The conditional mean grows from 0 to 9 and the conditional standard deviation from 1 to 4 over $[0, 3]$, giving a $16\times$ increase in variance. Observations are sorted by x before all subsequent steps. A full analysis is given in Appendix F.

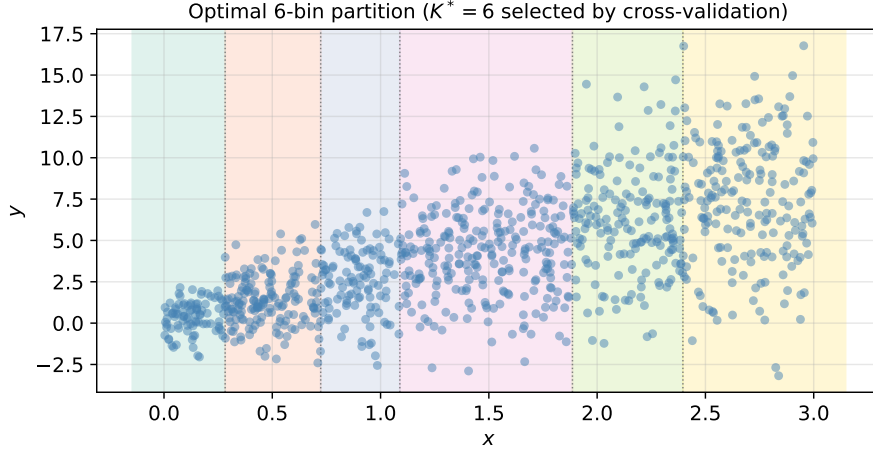


Figure 2: The synthetic dataset with the optimal 6-bin partition ($K^* = 6$). Shaded regions correspond to the six bins; dotted vertical lines mark the bin boundaries (midpoints between adjacent training observations).

8. Predictive Distributions: Conformal Inference

Having selected K^* and fitted the partition on all data, each test point x^* falls in some bin B_k with m training observations $y_1, \dots, y_m$. The partition enables two complementary predictive outputs: a *Venn prediction band*—a constant-width family of augmented ECDFs that is valid as a multiprobabilistic prediction under exchangeability (derivation and figure in Appendix D)—and a *conformal prediction set*, which adapts to local density and is the focus of this section.

8.1. Conformal Prediction Set

We briefly recall the key concepts of conformal prediction (Vovk et al., 2022) in the context of our setting and we show how to obtain valid prediction sets.

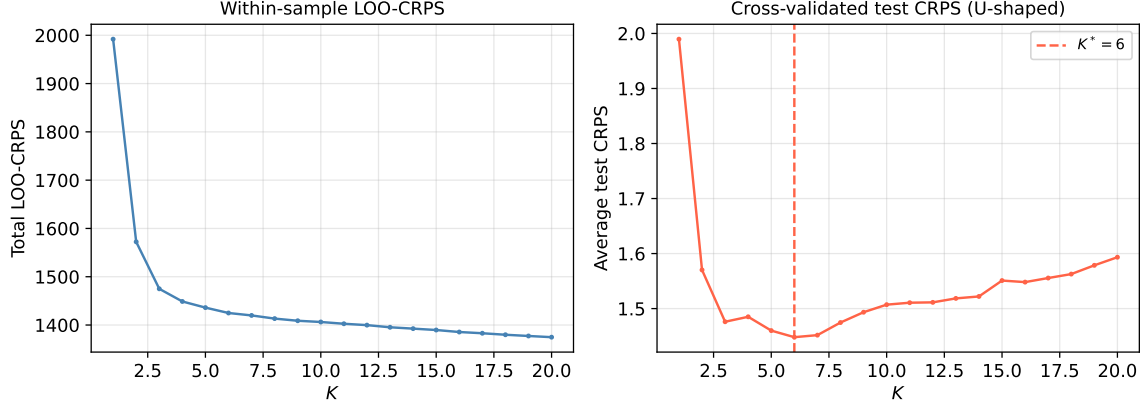


Figure 3: Within-sample LOO-CRPS (left) and cross-validated test CRPS (right) as functions of K on the running example. The within-sample criterion is nearly monotone decreasing, confirming its susceptibility to optimism bias (Hastie et al., 2009). The test CRPS has a clear U-shape with minimum at $K^* = 6$. Note that the left and right y -axis scales differ: the left shows a total (sum over all observations), the right an average (per held-out test point).

Nonconformity score. For test candidate y_h , define

$$\alpha(y_h) = \text{CRPS}(\hat{F}_m, y_h) = \frac{1}{m} \sum_{i=1}^m |y_i - y_h| - \frac{W}{m^2},$$

where $W = \sum_{i < j} |y_i - y_j|$ is the pairwise dispersion of the training bin (independent of y_h). For each training observation y_j ($j = 1, \dots, m$) in the augmented set $\{y_1, \dots, y_m, y_h\}$, define the leave-one-out nonconformity score

$$\alpha_j(y_h) = \text{CRPS}(F_{m+1}^{(-j)}, y_j),$$

where $F_{m+1}^{(-j)}$ is the ECDF of $\{y_1, \dots, y_m, y_h\} \setminus \{y_j\}$. Write $\alpha_{m+1}(y_h) \equiv \alpha(y_h)$ for the test score.

Conformal p-value.

$$p(y_h) = \frac{1}{m+1} \#\{j \in \{1, \dots, m+1\} : \alpha_j(y_h) \geq \alpha(y_h)\}.$$

Prediction set.

$$\Gamma^\varepsilon = \{y_h \in \mathbb{R} : p(y_h) > \varepsilon\}.$$

Proposition 2 *Under exchangeability of $(y_1, \dots, y_m, y^*)$, $\Pr(y^* \in \Gamma^\varepsilon) \geq 1 - \varepsilon$.*

For a proof, refer to the standard conformal prediction literature (Vovk et al., 2022).

Remark on exchangeability. A data-dependent partition mildly violates the exchangeability condition underlying Proposition 2: the DP uses all y -values to set boundaries, so conditioning on a bin’s membership acts as a y -dependent filter. The violation is less severe as bin sizes grow, and Theorem 2.1 in (Allen et al., 2025) provide formal support: auto-calibrated within-bin predictors inherit conformal guarantees even for data-dependent partitions. Further discussion on exchangeability is in Appendix E.

The suitability of LOO-CRPS as a nonconformity score Several properties make the LOO-CRPS a natural and principled nonconformity measure in this setting.

Properness implies sensitivity to genuine nonconformity. Because CRPS is strictly proper, the expected score $\mathbb{E}_P[\text{CRPS}(\hat{F}, Y)]$ is uniquely minimised when $\hat{F} = P$. Consequently, $\alpha(y_h) = \text{CRPS}(\hat{F}_m, y_h)$ is large precisely when y_h is surprising under the within-bin training distribution, not merely when it is far from some summary statistic such as the mean or median. Nonconformity scores based on absolute residuals $|y_h - \hat{\mu}|$ (or quantile residuals) implicitly assume a parametric location or quantile model for the bin; CRPS makes no such assumption, which matters when the within-bin distribution is skewed or multimodal.

The LOO structure guarantees exchangeability. Conformal validity requires that the scores $\alpha_1(y_h), \dots, \alpha_m(y_h), \alpha(y_h)$ be *exchangeable* under exchangeability of $(y_1, \dots, y_m, y^*)$. This holds here because every score is computed in exactly the same way: each α_j is the CRPS of the m -atom ECDF of the remaining m elements of $\{y_1, \dots, y_m, y_h\}$ evaluated at y_j , and $\alpha(y_h)$ is the same quantity for the test element. If instead we used the full-sample $\hat{F}_m$ (without the LOO adjustment) to score both training and test points, the training scores $\text{CRPS}(\hat{F}_m, y_j)$ and the test score $\text{CRPS}(\hat{F}_m, y^*)$ would not be on equal footing: the training ECDF was fitted using y_j , but not using y^* , breaking the symmetry on which conformal coverage rests.

Coherence with the binning criterion. The DP selects bin boundaries by minimising total LOO-CRPS on the training set. The conformal step then evaluates the test point by the same quantity: the marginal LOO-CRPS of adding y^* to the selected bin. This coherence means the bin is already optimised for the task of distributional prediction, and the prediction set Γ^ε has the natural interpretation as the set of test values whose inclusion would not increase the bin’s per-observation LOO-CRPS beyond the level seen at the training points (see also Section 8.2). Figure 4 illustrates the p-value curve at three test locations.

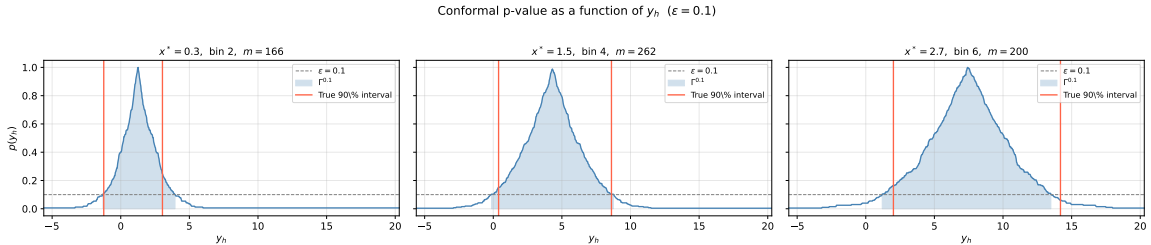


Figure 4: Conformal p-value $p(y_h)$ as a function of the candidate response y_h at three test points $x^* \in \{0.3, 1.5, 2.7\}$ for $\varepsilon = 0.10$. The shaded region is the prediction set $\Gamma^{0.10}$; vertical red lines mark the true 90% interval under $\mathcal{N}(3x^*, (1 + x^*)^2)$. The p-value curve is unimodal (each monotone piece is convex, since the underlying nonconformity score $\alpha(y_h)$ is convex), yielding a single connected prediction set in each case.

Computational advantages. CRPS appears unique among strictly proper scoring rules in admitting a closed-form LOO sum computable in $O(\log n)$ per bin extension; log score and Brier score are not known to admit analogous closed forms, and their straightforward evaluation would require $O(m \log m)$ or more per bin extension, compromising the $O(n^2 \log n)$ precomputation. CRPS also serves double duty as the conformal nonconfor-

mity score, aligning bin selection with prediction-set construction. A reformulation of the cross-validated criterion in terms of the Cramér distance—which generalises CRPS to distributional observations—is given in Appendix G.

8.2. Connection to the DP Cost

The sum of all $m+1$ nonconformity scores in the augmented set equals the total LOO-CRPS of that set:

$$\sum_{j=1}^{m+1} \alpha_j(y_h) = \text{cost}(\{y_1, \dots, y_m, y_h\}) = \frac{m+1}{m^2} W(\{y_1, \dots, y_m, y_h\}).$$

The test score $\alpha(y_h) = \text{CRPS}(\hat{F}_m, y_h)$ measures how much y_h increases the mean absolute deviation from the bin; it is large when y_h is far from the bulk of $y_1, \dots, y_m$. The prediction set Γ^ε therefore consists of values whose addition would not dramatically inflate the DP cost of bin B_k .

Figure 5 shows the resulting prediction intervals across the covariate range.

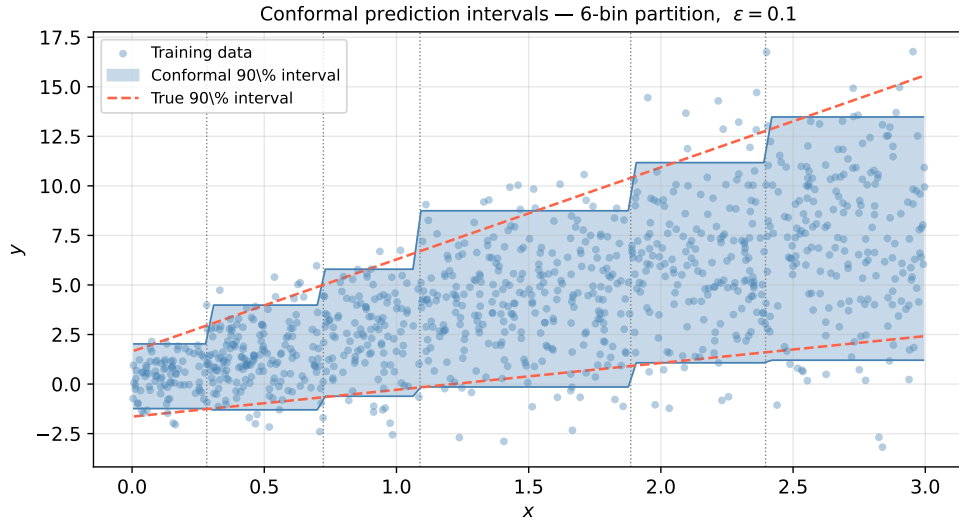


Figure 5: Conformal 90% prediction intervals (shaded blue) and true 90% intervals (dashed red) across the covariate range, with bin boundaries marked by dotted vertical lines. The interval width adapts to the within-bin spread, tracking the increasing conditional variance of the heteroscedastic data-generating process.

8.3. Interval Structure and Non-Convex Extensions

Γ^ε is approximately a single interval. The convexity of the CRPS nonconformity score is not incidental; it is a structural consequence of measuring average distance to the training distribution. Writing

$$\alpha(y_h) = \frac{1}{m} \sum_{i=1}^m |y_i - y_h| - \frac{W}{m^2},$$

the second term is constant in y_h , and the first is a sum of convex functions $|y_i - y_h|$, hence convex with a unique minimum at the empirical median of $\{y_1, \dots, y_m\}$. More generally, any score of the form $\mathbb{E}_{\hat{F}_m}[\ell(y_h, Y)]$ with ℓ convex in its first argument (e.g. squared error, absolute error, CRPS) inherits this property. Consequently, the sublevel set $\{y_h : \alpha(y_h) \leq c\}$ is a closed interval for every $c \geq 0$.

In a split-conformal predictor the training scores do not depend on y_h , so $p(y_h)$ is non-increasing in $\alpha(y_h)$ and convexity of α directly implies that Γ^ε is a connected interval. In the present transductive (full-data) setting all $m + 1$ scores depend on y_h , and their relative ranking can in principle change as y_h varies, so the split-conformal argument does not apply directly. We conjecture (and, to be clear, this remains an open problem) that Γ^ε remains a connected interval whenever $\alpha(y_h)$ is convex; in all our experiments (synthetic, Old Faithful, motorcycle; m ranging from 22 to 262) the prediction set is a single interval without exception.

The consequence for multimodal bins is concrete. Suppose the training responses in B_k are bimodal with well-separated modes at $\mu_1 \ll \mu_2$. The empirical median lies in the inter-modal gap; $\alpha(y_h)$ attains its minimum there. The resulting Γ^ε spans both modes and the low-density gap between them: it is valid (marginal coverage is guaranteed) and correctly wide, but informationally wasteful as it assigns coverage to a region of near-zero probability mass.

A bandwidth-free non-convex alternative: the k -NN score. Non-contiguous prediction sets require a nonconformity score that is non-convex in y_h , i.e. one that is simultaneously small near each mode and large in the inter-modal gap. Density-based scores such as $-\log \hat{f}(y_h)$ achieve this but require a bandwidth. In one dimension, the k -nearest-neighbour (k -NN) distance provides a bandwidth-free alternative. Define

$$\alpha^{(k)}(y_h) = d_{(k)}(y_h, \{y_1, \dots, y_m\}),$$

the k -th smallest value of $|y_h - y_i|$ over $i = 1, \dots, m$. For $k = 1$ this is simply $\min_i |y_h - y_i|$, which has a local minimum at every training point and is large precisely where the training distribution is sparse.

The LOO version for the conformal construction is defined symmetrically in the augmented set $\{y_1, \dots, y_m, y_h\}$: for each training observation y_j ,

$$\alpha_j^{(1)}(y_h) = \min\left(\min_{i \neq j} |y_j - y_i|, |y_j - y_h|\right),$$

and $\alpha_{m+1}^{(1)}(y_h) \equiv \alpha^{(1)}(y_h)$. Under exchangeability of $(y_1, \dots, y_m, y^*)$, if we define as the conformal p-value

$$p^{(1)}(y_h) = \frac{1}{m+1} \#\{j : \alpha_j^{(1)}(y_h) \geq \alpha^{(1)}(y_h)\}$$

then Proposition 2 holds without modification.

The prediction set $\Gamma^{\varepsilon, (1)} = \{y_h : p^{(1)}(y_h) > \varepsilon\}$ is approximately the union

$$\bigcup_{i=1}^m [y_i - c_\varepsilon, y_i + c_\varepsilon],$$

where c_ε is the $(1 - \varepsilon)$ -quantile of the training LOO scores (exact in the limit $m \rightarrow \infty$ where the y_h -dependence of $\alpha_j(y_h)$ vanishes). For a bimodal distribution, training points cluster near both modes, so c_ε is set by the intra-cluster spacing; if this is smaller than half the inter-modal gap, the prediction set consists of two disjoint intervals.

The effective resolution is determined purely by the data; no bandwidth is specified by the user.² The conformal threshold c_ε plays the role of a density threshold, automatically calibrated for $1 - \varepsilon$ coverage. The resulting $\Gamma^{\varepsilon, (1)}$ approximates a non-parametric highest-density region (HDR) of the within-bin empirical distribution.

Binning remains LOO-CRPS optimal. The k -NN modification applies only to the conformal step; the DP binning continues to use the LOO-CRPS cost of Section 4. We argue that this separation is principled: the binning objective is to group covariate-sorted observations so that the within-bin ECDF is a good predictive distribution for the local conditional $P(Y \mid X = x)$, which is a question of predictive accuracy for which LOO-CRPS is a natural criterion regardless of whether the within-bin distribution is unimodal or multimodal.

Using $\sum_{j \in B_k} \alpha_j^{(1)}$ as the DP cost would be computationally feasible (the cost is additive over bins, so optimal substructure is preserved), but it would measure local y -density rather than predictive quality, and it would exhibit the same optimism bias as within-sample LOO-CRPS: a size-2 bin with $|y_1 - y_2| \approx 0$ has near-zero 1-NN cost, driving the DP to over-partition. The two steps therefore use different criteria for different purposes: LOO-CRPS for reference-class selection, and either CRPS or k -NN for conformal evaluation.

Synthetic illustration. Figure 6 illustrates the contrast on a full-regression bimodal mixture DGP with $n = 600$ training observations: $Y \mid X=x \sim 0.5\mathcal{N}(x-1.5, 0.3^2) + 0.5\mathcal{N}(x+1.5, 0.3^2)$, $X \sim \text{Uniform}(0, 6)$. The two conditional modes run as parallel tracks separated by 3 units across the covariate range. Cross-validation selects $K^* = 6$ bins; both panels use the same partition. The *left* panel shows CRPS conformal intervals: for each bin the prediction set is a single wide rectangle spanning both mode tracks and the low-density gap between them. The *right* panel shows k -NN conformal intervals with $k^* = 7$, selected by cross-validating average prediction-set length: for each bin the set narrows to intervals tracking each mode track, excluding the inter-modal gap. Both scores carry the coverage guarantee; the visible narrowing in the right panel corresponds to the efficiency gain from matching the prediction set to the within-bin conditional support.

8.4. Approximate Exchangeability and the Bias–Variance Trade-off in K

Wide bins (small K) average over a heterogeneous region, hurting exchangeability; narrow bins (large K) improve exchangeability but yield coarser intervals, since the conformal p-value grid is $\{1/(m+1), \dots, 1\}$. The CV criterion of Section 7 penalises both failure modes and yields a genuine U-shaped optimum (see Appendix E for the full analysis); K is effectively the method’s only tuning parameter, since the construction imposes no fixed functional form on $P(Y \mid X = x)$.

2. It was pointed out to the author that indeed this adaptivity is rooted in a classical result from one-dimensional order statistics: the 1-NN distance is a consistent density estimator without smoothing, since $m \cdot \min_i |y_h - Y_i|$ converges in distribution to $\text{Exp}(f(y_h))$ when $Y_1, \dots, Y_m \sim f$ (Devroye and Györfi, 1985).

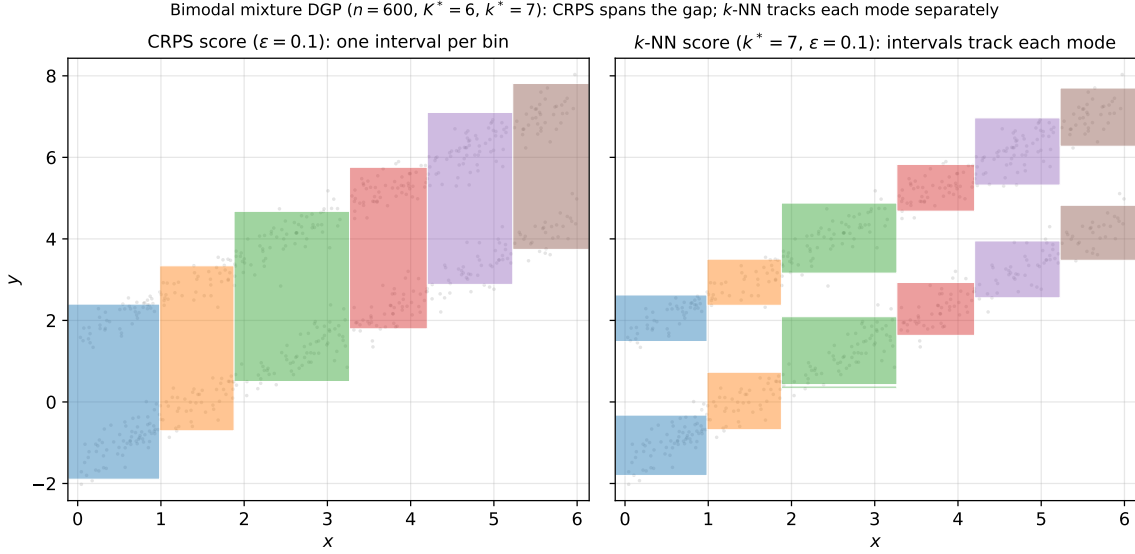


Figure 6: Bimodal mixture DGP ($n = 600$, $\varepsilon = 0.10$, $K^* = 6$). Shaded rectangles show the conformal prediction set for each bin; grey dots are training observations. *Left*: CRPS score — one wide interval per bin, spanning both mode tracks. *Right*: k -NN score ($k^* = 7$, CV-selected) — narrow intervals tracking each mode; the low-density inter-modal gap is excluded.

9. Real-Data Experiments

We evaluate the method on two real datasets that stress different aspects of distributional non-stationarity: a bimodal conditional distribution and a strongly heteroscedastic one. All experiments use $\varepsilon = 0.10$ (nominal 90% coverage). Competitors are evaluated over $R = 200$ random 50/50 train/calibration splits, with standard errors reported throughout.

Competitors.

- **Gaussian split conformal.** OLS fit on the training half; absolute residuals on the calibration half determine the constant-width prediction interval.
- **CQR (cubic).** Conformalized Quantile Regression (Romano et al., 2019) with cubic polynomial base quantile regressors ($\tau = 0.05$ and 0.95), calibrated on the same held-out half.
- **CQR-QRF.** Quantile Regression Forest (Meinshausen, 2006) with 500 trees fitted on the training half, conformalized with the $\max(q_{\alpha/2}(x) - y, y - q_{1-\alpha/2}(x))$ score (same as CQR) calibrated on the held-out half.
- **CQR-IDR.** Isotonic Distributional Regression (Henzi et al., 2021) fitted on the training half, with quantile predictions at levels $\alpha/2$ and $1 - \alpha/2$ conformalized via the same CQR score on the calibration half.

Gaussian split conformal and CQR (cubic) are parametric (fixed functional form); CQR-QRF, CQR-IDR, QOOB (Gupta et al., 2022), and our method are non-parametric, as is CHR here via its QRF black box (Appendix H) rather than the framework itself. Our

method is transductive: all n observations are used for both partitioning and conformal calibration (full-data conformal with the within-bin ECDF as the nonconformity score), with no data held out. Split-conformal competitors must reserve a calibration set, effectively halving the sample available for model fitting. This data-efficiency advantage is inherent to full-data conformal methods (Barber et al., 2021). To disentangle the effect of the method itself from this sample-size advantage, each table below reports two rows for our method: the *full- n* row reflects the method as deployed (transductive, no data splitting), while the *$n/2$* row handicaps it to the same training set available to competitors, providing a controlled comparison.

9.1. Old Faithful: bimodal conditional distribution

The **faithful** dataset ($n = 272$) records eruption duration and waiting time between eruptions of the Old Faithful geyser. We pick $x =$ waiting time (min) as the predictor and $y =$ eruption duration (min) as the response. The marginal distribution of eruption duration is bimodal: short eruptions (≈ 2 min) and long eruptions (≈ 4.5 min), with the mixture proportion shifting as a function of waiting time. This is a natural stress-test for methods that assume a unimodal conditional distribution.

Cross-validation selects $K^* = 4$, with bin boundaries at waiting times 63.0, 67.5, and 71.5 minutes. Figure 7 (left) shows the partition; Figure 7 (right) shows the within-bin ECDFs. The two outer bins have clearly unimodal within-bin distributions (short eruptions for short waits, long eruptions for long waits), while the two inner bins capture the transition region with finer resolution.

Figure 8 compares 90% prediction intervals across methods. Gaussian split conformal and CQR produce intervals of roughly constant width across all waiting times, whereas our method adapts: narrower in both regimes where the conditional distribution is concentrated, and wide only in the transition region where the within-bin ECDF spans both modes.

Table 1 reports coverage and mean interval width averaged over $R = 200$ random 50/50 splits. In the matched-sample comparison (top block), our method (*$n/2$* , italic row) achieves mean width 1.27 min with slightly sub-nominal coverage ($88.5 \pm 0.3\%$), narrower than all competitors: Gaussian split conformal (1.68 min), CQR (1.49 min), CQR-IDR (1.29 min), CQR-QRF (1.33 min), and CHR (Sesia and Romano, 2021) (1.31 min). CHR, reimplemented independently from the published algorithm for this comparison (Appendix H documents the implementation and benchmarking choices in detail), is narrower than CQR-QRF despite using the identical underlying quantile-regression forest (1.312 ± 0.009 min vs. 1.333 ± 0.006 min) at closely comparable coverage. When all n observations are used (bold row, below the line), our method achieves $90.6 \pm 0.1\%$ coverage with mean width 1.22 min. All split-conformal methods exceed nominal coverage (90.6–91.4%). Among full-data transductive competitors, QOOB (Gupta et al., 2022) achieves a narrower interval still (1.157 ± 0.001 min at $92.5 \pm 0.1\%$ coverage) — about 5% narrower than our method at comparable coverage, confirming that the transductive efficiency advantage reported here is not unique to this specific partition-based construction.

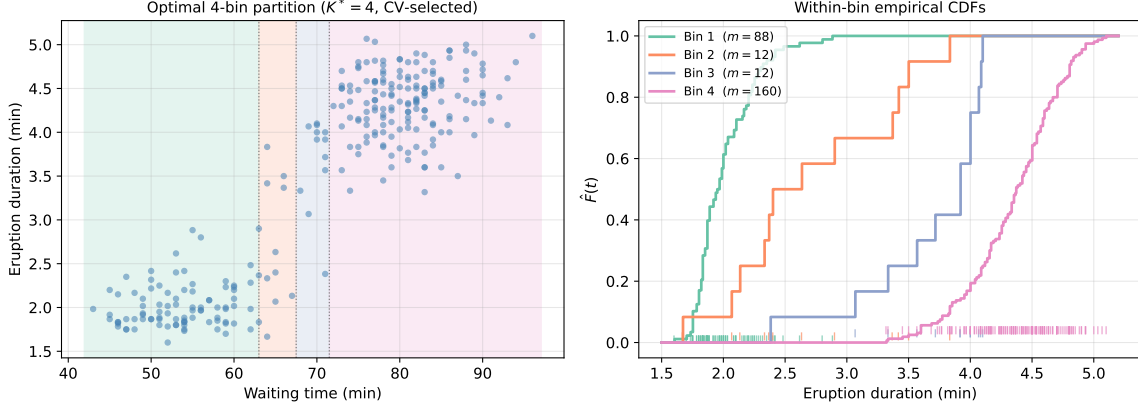


Figure 7: Old Faithful. *Left*: scatter of eruption duration vs. waiting time with the optimal 4-bin partition. Boundaries at 63.0, 67.5, and 71.5 minutes resolve the transition between the short-eruption and long-eruption regimes. *Right*: within-bin empirical CDFs; the outer bins are unimodal, confirming the partition captures the regime structure.

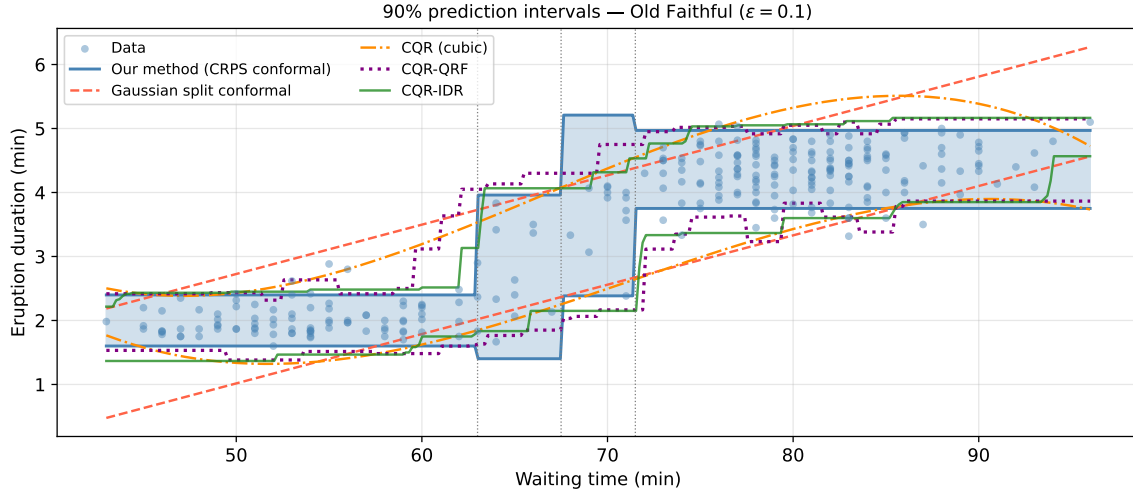


Figure 8: Old Faithful: 90% prediction intervals from five methods. Our CRPS-based conformal intervals adapt to the local conditional spread within each bin, producing narrower intervals than all competitors in both regimes. In the transition region around 67–70 minutes, the proposed method’s interval is deliberately wide: the single bin spanning both eruption modes must cover the full bimodal spread, an honest consequence of the partition geometry.

9.2. Motorcycle accident: heteroscedastic benchmark

The `mcycle` dataset ($n = 133$) records head acceleration of a motorcycle dummy as a function of time after a simulated impact. The response variance changes dramatically: near-zero before ≈ 15 ms, explosive in the 15–30 ms deformation phase, and moderating thereafter. This is the standard benchmark for heteroscedastic prediction intervals in the nonparametric regression literature.

Cross-validation selects $K^* = 6$, with boundaries at 15.1, 17.6, 24.4, 27.2, and 38.0 ms, clustered in the high-variance impact region. Figure 9 shows the K-selection curve

Method	Coverage (%)	Mean width (min)
<i>Our method ($n/2$)</i>	88.5 ± 0.3	1.270 ± 0.010
Gaussian split conformal	91.2 ± 0.0	1.683 ± 0.006
CQR (cubic)	91.2 ± 0.0	1.490 ± 0.006
CQR-QRF	91.4 ± 0.0	1.333 ± 0.006
CQR-IDR	91.4 ± 0.0	1.294 ± 0.007
CHR (Sesia and Romano, 2021)	90.6 ± 0.1	1.312 ± 0.009
Our method (full n)	90.6 ± 0.1	1.217 ± 0.002
QOOB (full n) (Gupta et al., 2022)	92.5 ± 0.1	1.157 ± 0.001

Table 1: Old Faithful ($n = 272$): empirical coverage and mean width of prediction intervals at nominal level $1 - \varepsilon = 0.90$, averaged over $R = 200$ random 50/50 splits ($\pm$ one standard error). *Top block*: all methods use the same training half ($n/2$); *Our method ($n/2$)* is directly comparable to the competitors; CHR (Sesia and Romano, 2021) uses the same quantile-regression-forest grey-box as CQR-QRF (Appendix H). *Below the line*: **Our method (full n)** uses all n observations, a design advantage inherent to full-data conformal methods; QOOB (Gupta et al., 2022) is a different transductive (full-data) method, included as an additional comparison at the same data-usage regime.

and the resulting partition. With $n = 133$ observations, $K^* = 6$ implies bins of roughly 10–30 observations; the narrower bins in the chaotic 15–30 ms window improve within-bin exchangeability at the cost of fewer ECDF atoms. This is the small-sample manifestation of the bias–variance trade-off discussed in Section 8.4: the CV criterion selects the best available tradeoff, not a guaranteed minimum bin size, and the coarseness of the within-bin ECDF is absorbed into interval width rather than miscoverage.

Figure 10 and Table 2 compare prediction intervals averaged over $R = 200$ random 50/50 splits. In the matched-sample comparison (top block), Gaussian split conformal is drastically inefficient (172.4 g), CQR (cubic) and CQR-IDR are moderately better (134.1 g and 127.6 g respectively), and CQR-QRF and CHR (Sesia and Romano, 2021) adapt most flexibly (87.9 g and 84.5 g respectively); as on Old Faithful, CHR is narrower than CQR-QRF with an identical grey-box (Appendix H), here also at closer-to-nominal coverage ($91.3 \pm 0.2\%$ vs. $93.1 \pm 0.1\%$). Our method on the same training half (italic row, $n_{\text{tr}} = 66$) yields mean width 100.6 g with sub-nominal coverage ($86.9 \pm 0.5\%$); at this small sample size, CQR-QRF achieves narrower intervals, reflecting the bias–variance trade-off discussed in Section 8.4: fewer observations per bin produce a coarser within-bin ECDF and wider intervals. When all n observations are used (bold row, below the line), our method achieves $90.3 \pm 0.2\%$ coverage with mean width 77.3 g, narrower than all competitors. All split-conformal methods exceed nominal coverage (91.3–93.1%). Among full-data transductive competitors, QOOB (Gupta et al., 2022) is comparable (75.8 $\pm$ 0.2 g at $92.5 \pm 0.2\%$ coverage) — a similar width achieved with a larger coverage margin; on this dataset 15% of QOOB’s prediction sets are non-convex (up to 10 disjoint segments), unlike our method’s single-interval sets observed throughout.

10. Additional considerations

Limitations. In terms of scalability, it is important to note that the DP binning step has $O(n^2K)$ time complexity and $O(n^2)$ storage requirement, which may be prohibitive for

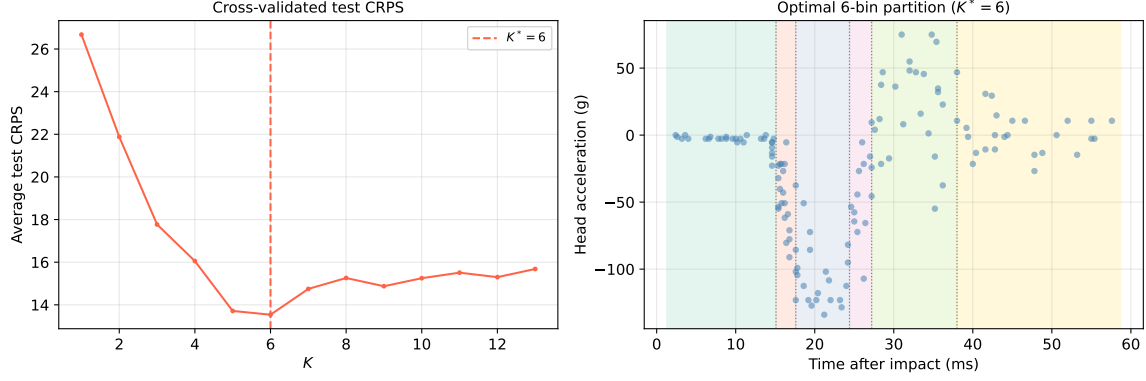


Figure 9: Motorcycle accident. *Left*: cross-validated test CRPS as a function of K with the selected $K^* = 6$. *Right*: scatter with the 6-bin partition; boundaries are concentrated in the high-variance impact phase.

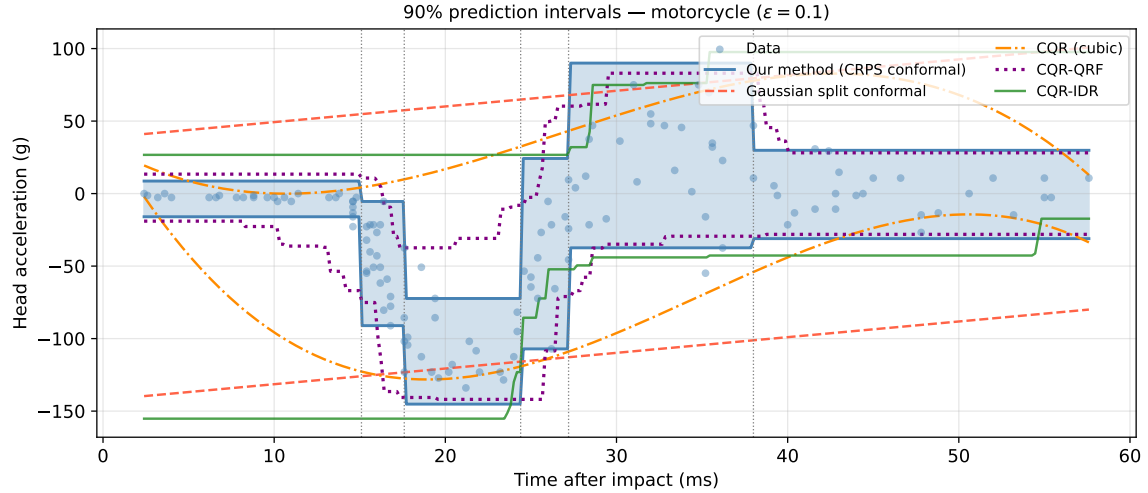


Figure 10: Motorcycle accident: 90% prediction intervals. Gaussian split conformal is constrained to constant width; our method, CQR-QRF, and CQR-IDR adapt to the non-stationary variance structure. A small number of data points in the high-variance impact phase (20–30 ms) fall outside the proposed method’s intervals, consistent with the approximate within-bin exchangeability in this small and strongly heteroscedastic dataset.

large datasets. The method applies only to the case of a one-dimensional covariate and a contiguous binning structure. In small samples the CV criterion can select a large K , resulting in bins with few observations and hence coarse prediction intervals; this is a feature of the bias-variance tradeoff under limited data, not a failure of validity. The method also assumes approximate local homogeneity of the conditional distribution within each bin; Section 8.4 and Appendix E analyse the resulting bias–variance trade-off in K , and the Old Faithful comparison against QOOB (Section 9) illustrates a real, if modest, efficiency cost when a smooth transition is approximated by hard bin boundaries.

Method	Coverage (%)	Mean width (g)
<i>Our method</i> ($n/2$)	86.9 ± 0.5	100.6 ± 1.3
Gaussian split conformal	92.5 ± 0.0	172.4 ± 1.0
CQR (cubic)	92.5 ± 0.0	134.1 ± 1.5
CQR-QRF	93.1 ± 0.1	87.9 ± 0.7
CQR-IDR	92.7 ± 0.0	127.6 ± 0.6
CHR (Sesia and Romano, 2021)	91.3 ± 0.2	84.5 ± 1.0
Our method (full n)	90.3 ± 0.2	77.3 ± 0.2
QOOB (full n) (Gupta et al., 2022)	92.5 ± 0.2	75.8 ± 0.2

Table 2: Motorcycle accident ($n = 133$): empirical coverage and mean width of prediction intervals at nominal level $1 - \varepsilon = 0.90$, averaged over $R = 200$ random 50/50 splits ($\pm$ one standard error). *Top block*: all methods use the same training half ($n/2$); *Our method* ($n/2$) is directly comparable to the competitors; CHR (Sesia and Romano, 2021) uses the same quantile-regression-forest grey-box as CQR-QRF (Appendix H). *Below the line*: **Our method** (full n) uses all n observations, a design advantage inherent to full-data conformal methods; QOOB (Gupta et al., 2022) is a different transductive (full-data) method, included as an additional comparison at the same data-usage regime.

Multi-dimensional extension. The LOO-CRPS cost $W(S) \cdot m / (m-1)^2$ can in principle be extended to multivariate data, but DP tractability relies on the total order induced by sorting a one-dimensional x . One could think of two natural extensions for $d > 1$.

The first is projection onto a learned one-dimensional index $g : \mathbb{R}^d \rightarrow \mathbb{R}$, sorting by $g(x)$ and running the existing DP. The partition is globally optimal *given* g ; alternating optimisation over $(g, \text{partition})$ is tractable, but global optimality in g is not guaranteed. An open question is whether the data-adaptive g minimising LOO-CRPS recovers a form of sufficient dimension reduction for the conditional *distribution* $P(Y | X = x)$, rather than merely the conditional mean.

The second is CART-style greedy binary splitting, requiring $O(dn)$ evaluations per split and $O(Kdn)$ total. No projection is needed and regions are interpretable rectangular cells, but global optimality is again not guaranteed.

Open problems. At least three questions remain unresolved. First, whether CRPS coherence (same rule for binning and scoring) can be proved to yield tighter coverage bounds or more efficient prediction sets relative to a mismatched pair. Second, the monotone-coverage question: as K increases, conditional coverage at a fixed x^* should improve; making this precise requires bounding the exchangeability violation as a function of bin width and local DGP smoothness. Third, the consistency of K^* : does $K^* = \arg \min_K \overline{\text{TestCRPS}}(K)$ converge to a meaningful oracle as $n \rightarrow \infty$, and under what conditions does the population $\overline{\text{TestCRPS}}(K)$ have a unique minimum?

11. Conclusion

We have presented a non-parametric, model-free method for conditional distribution estimation that combines three independently motivated ideas into a coherent pipeline: optimal bin-boundary placement by LOO-CRPS minimisation, cross-validated bin-count selection, and conformal prediction using the within-bin ECDF as both the predictive distribution and the nonconformity score.

The central technical contribution is the closed-form cost $\text{cost}(S) = mW/(m-1)^2$, which reduces the LOO-CRPS of any bin to a single pairwise-dispersion scalar and enables exact globally optimal partitioning in $O(n^2K)$ time via dynamic programming.

On the predictive side, convexity of the CRPS nonconformity score guarantees contiguous prediction intervals in the split-conformal case; in our transductive setting, single-interval structure is observed in all experiments. The k -NN score provides a bandwidth-free alternative that yields non-contiguous highest-density regions for multimodal within-bin distributions. The two scores are therefore complementary: CRPS for efficiency in unimodal regimes, k -NN for distributional fidelity when the within-bin response is multimodal.

On the real datasets, the full- n method (which uses all observations, unlike competitors that reserve half for calibration) is competitive or superior in interval efficiency: 11–40% narrower intervals on Old Faithful (bimodal conditional) and $2.2\times$ narrower than Gaussian split conformal on the motorcycle benchmark (strongly heteroscedastic), while maintaining near-nominal coverage in both cases. A matched-sample comparison (restricting our method to the same training half as the competitors) shows the cost of halving the data: on Old Faithful the method remains narrower than all competitors; on the motorcycle dataset ($n/2 = 66$, coverage $86.9 \pm 0.5\%$) CQR-QRF achieves modestly narrower intervals (87.9 vs 100.5 g), reflecting the small-sample bias-variance regime. Against QOOB (Gupta et al., 2022), another transductive full-data method, the picture is mixed: QOOB is $\sim 5\%$ narrower on Old Faithful at comparable coverage, and closely comparable on the motorcycle dataset — evidence that the efficiency gains reported above stem from this specific closed-form partitioning construction rather than from transductive calibration per se.

The margin over the two parametric baselines reflects dataset mismatch, not a general property of the method; the fairer test is against the non-parametric competitors, where the results above are mixed — notably with only K as a tuning parameter, against considerably more complex alternatives.

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Software and Reproducibility

The method is implemented in the `crpsconfreg` Python package, available at <https://pypi.org/project/crpsconfreg>. All figures and numerical results in this paper can be reproduced using the scripts in the accompanying repository at <https://github.com/ptocca/crps-conformal-regression>. An interactive browser demo (no installation required) is hosted at <https://ptocca.github.io/crps-conformal-regression/>.

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Appendix A. Proofs

Proof of Proposition 1

Sum the per-observation expression over $k \in S$. For the first term,

$$\sum_{k=1}^m \frac{d_k}{m-1} = \frac{1}{m-1} \sum_{k=1}^m d_k = \frac{D}{m-1},$$

since $\sum_k d_k = \sum_k \sum_{\ell \neq k} |y_\ell - y_k| = D$. For the second term, note that removing k from the pairwise sum gives $\sum_{\ell \neq r, \ell, r \neq k} |y_\ell - y_r| = D - 2d_k$, so

$$\sum_{k=1}^m \frac{D - 2d_k}{2(m-1)^2} = \frac{mD - 2D}{2(m-1)^2} = \frac{D(m-2)}{2(m-1)^2}.$$

Subtracting and simplifying,

$$\text{cost}(S) = \frac{D}{m-1} - \frac{D(m-2)}{2(m-1)^2} = \frac{mD}{2(m-1)^2} = \frac{mW}{(m-1)^2}.$$

□

Appendix B. Algorithm Pseudocode and Implementation Details

Pseudocode. The pseudocode in Algorithm 1 on page 24 implements the DP algorithm for computing the CRPS-optimal K -partition.

Choice of data structure. Two Fenwick trees (Fenwick, 1994) suffice: one storing counts (to obtain r) and one storing values (to obtain $S_\leq$). A sorted array answers prefix queries in $O(1)$ but requires $O(m)$ per insertion, raising total precomputation to $O(n^3)$. A balanced BST achieves the same $O(\log n)$ per operation with higher constant factors; the Fenwick tree is therefore the natural choice: simple, cache-friendly, and sufficient for the $O(n^2 \log n)$ bound.

Appendix C. Tightness of $O(n^2 K)$: Quadrangle Inequality Counterexample

If $c(i, j)$ satisfied the *quadrangle inequality* (QI), $c(a, c) + c(b, d) \leq c(a, d) + c(b, c)$ for all $a \leq b \leq c \leq d$, then by the Knuth–Yao theorem (Knuth, 1971; Yao, 1980) optimal split points would be monotone in j , enabling $O(n)$ per DP row and total $O(nK)$. However, the LOO-CRPS cost violates the QI already at $n = 4$: for $\mathbf{y} = (0, 0, 0, 0, 0, 1, 1, 1, 1, 1)$ and $(a, b, c, d) = (1, 4, 5, 6)$ the inequality fails by a gap of 0.30. One could view this as a consequence of the $m/(m-1)^2$ prefactor which makes the cost disproportionately sensitive to small bins.

Appendix D. Venn Prediction Band

For test point y^* in bin B_k with m training responses $y_1, \dots, y_m$, the Venn prediction is the family of augmented ECDFs

$$F_h(t) = \frac{1}{m+1} \left(\#\{i : y_i \leq t\} + \mathbf{1}[y_h \leq t] \right), \quad y_h \in \mathbb{R}.$$

Each F_h is a valid CDF; the family spans all distributions consistent with one additional observation in the bin. At each t , the band covers

$$\underline{F}(t) = \frac{\#\{i : y_i \leq t\}}{m+1}, \quad \bar{F}(t) = \underline{F}(t) + \frac{1}{m+1},$$

Algorithm 1: CRPS-optimal K -partition via dynamic programming

Input: Sorted observations $(x_{(1)}, y_1), \dots, (x_{(n)}, y_n)$; number of bins K

Output: Optimal bin boundaries $0 = b_0 < b_1 < \dots < b_K = n$

Phase 1: Precompute cost matrix;

for $i = 1$ **to** n **do**

 Initialise Fenwick trees $T_{\text{cnt}}, T_{\text{sum}}$; running total $\Sigma \leftarrow 0$; $W \leftarrow 0$;

for $j = i$ **to** n **do**

 Insert y_j into T_{cnt} and T_{sum} ; update $\Sigma \leftarrow \Sigma + y_j$;

$r \leftarrow T_{\text{cnt}}.\text{prefixQuery}(y_j)$; $S_{\leq} \leftarrow T_{\text{sum}}.\text{prefixQuery}(y_j)$; $S_{>} \leftarrow \Sigma - S_{\leq}$;

$W \leftarrow W + y_j \cdot r - S_{\leq} + S_{>} - y_j \cdot (j - i + 1 - r)$;

$m \leftarrow j - i + 1$;

$c[i][j] \leftarrow \begin{cases} m W / (m - 1)^2 & \text{if } m \geq 2 \\ +\infty & \text{if } m = 1 \end{cases}$;

end

end

Phase 2: Fill DP table;

for $j = 2$ **to** n **do**

$\text{dp}[1][j] \leftarrow c[1][j]$; $\text{split}[1][j] \leftarrow 0$;

end

for $k = 2$ **to** K **do**

for $j = k$ **to** n **do**

$\text{dp}[k][j] \leftarrow +\infty$;

for $i = k - 1$ **to** $j - 1$ **do**

$v \leftarrow \text{dp}[k - 1][i] + c[i + 1][j]$;

if $v < \text{dp}[k][j]$ **then**

$\text{dp}[k][j] \leftarrow v$; $\text{split}[k][j] \leftarrow i$;

end

end

end

end

Phase 3: Backtrack boundaries;

$b_K \leftarrow n$; $k \leftarrow K$; $j \leftarrow n$;

while $k \geq 1$ **do**

$b_{k-1} \leftarrow \text{split}[k][j]$; $j \leftarrow b_{k-1}$; $k \leftarrow k - 1$;

end

return $(b_0, b_1, \dots, b_K)$;

a constant width of $1/(m+1)$ everywhere. In terms of the training ECDF $\hat{F}_m$: $\underline{F}(t) = \frac{m}{m+1}\hat{F}_m(t)$, $\overline{F}(t) = \underline{F}(t) + \frac{1}{m+1}$. Under exchangeability, this construction gives a valid multiprobabilistic prediction in the sense of Vovk et al.

Figure 11 shows the band for each bin of the running example. The width $1/(m+1)$ is determined entirely by bin size and carries no information about the local density or the position of y^* within the bin.

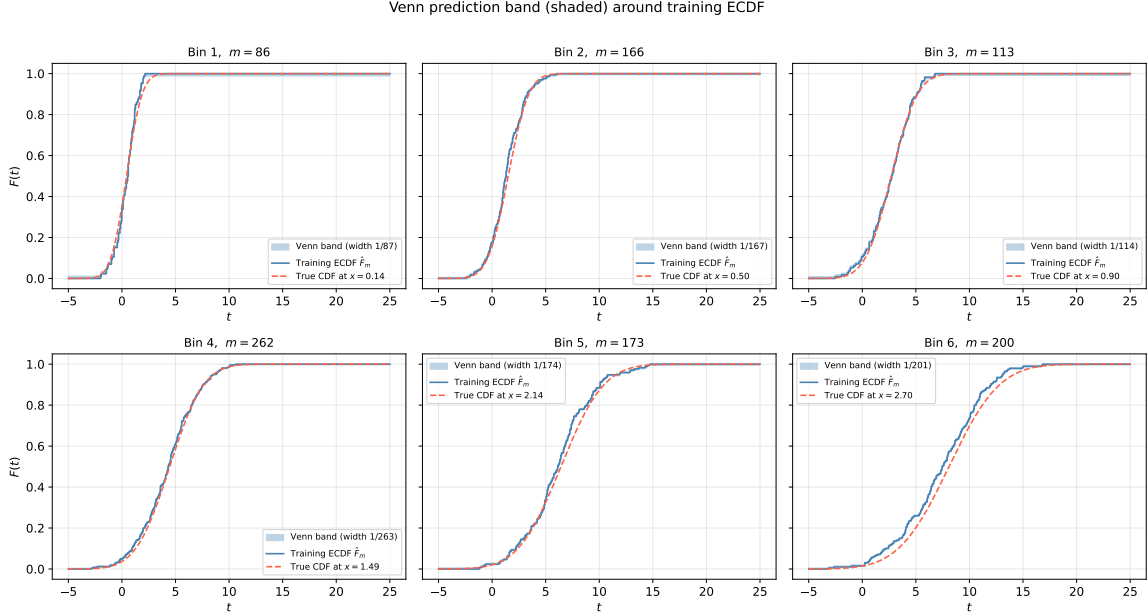


Figure 11: Venn prediction band (shaded) and training ECDF $\hat{F}_m$ (step function) for each of the six bins, alongside the true conditional CDF at the bin midpoint (dashed red). The band width $1/(m+1)$ ranges from 0.004 to 0.012, invisible at these bin sizes.

Appendix E. Exchangeability Analysis and Bias–Variance Trade-off

Exchangeability analysis

Conformal validity (Proposition 2) requires two conditions:

1. **Score symmetry.** For any fixed set of $m+1$ values in a bin, any permutation of $(y_1, \dots, y_m, y^*)$ produces the same multiset of LOO nonconformity scores. This is a property of the LOO-CRPS computation and always holds.
2. **Statistical exchangeability of bin members.** Under i.i.d. sampling, the $m+1$ values in the bin must be exchangeable as random variables, which requires that bin membership does not introduce dependence among the y -values.

Condition (2) is mildly violated by a data-dependent partition. The DP uses all y -values to set boundaries, so conditioning on a bin’s membership acts as a y -dependent filter: observations ended up together partly *because* their y -values were compatible with the DP’s cost criterion.

A concrete thought experiment makes this vivid. Consider an observation near a bin boundary. If its y -value were very different from its bin-mates, the DP might have placed the boundary on the other side. Conditioning on it being in *this* bin therefore biases its y -distribution toward values compatible with the present companions.

We argue informally that the violation is mild in practice: it is not in the score computation (permutation-invariant for any fixed bin) but in the probability model over which observations end up together. For the partition to change, a single observation must alter the DP cost matrix enough to shift a boundary — increasingly unlikely as bin sizes grow. [Allen et al. \(2025\)](#) provide formal support: auto-calibrated within-bin predictors inherit conformal guarantees even for data-dependent partitions.

Bias–variance trade-off in K

Wide bins (small K) average over a heterogeneous x -range, violating exchangeability; the resulting conformal intervals may be mis-sized at a specific x^* — conservative where local spread is below the bin average, anti-conservative where it is above.

Narrow bins (large K) improve exchangeability but at two distinct costs. First, the conformal p-value lies on the grid $\{1/(m+1), \dots, 1\}$, so for $m < \lceil 1/\varepsilon \rceil - 1$ no point can be excluded at level ε and the prediction set is the whole real line. At $\varepsilon = 0.10$ this floor falls at $m < 9$. Second, with small m the conformal quantile is estimated from few scores, so interval widths are highly variable across test points.

The CV criterion penalises both failure modes: over-split partitions incur high test CRPS because the within-bin ECDF has too few atoms, while under-split partitions incur high test CRPS because the ECDF averages over too heterogeneous a region. The U-shaped $\text{TestCRPS}(K)$ is therefore a genuine empirical optimum.

Appendix F. Numerical Illustration

We illustrate further the running example introduced in the main text in Section 7: $n = 1000$, $Y \mid X = x \sim \mathcal{N}(3x, (1+x)^2)$, $X \sim \text{Uniform}(0, 3)$.

Cross-validated K selection. The 5-fold CV procedure selects $K^* = 6$; the CV curves are shown in Figure 3 and the resulting partition in Figure 2. The bin boundaries and sizes are listed below.

Conformal prediction intervals. Table 3 reports the 90% conformal prediction intervals ($\varepsilon = 0.10$) at three representative test points, alongside the true 90% intervals under the generating distribution.

x^*	Bin	$\Gamma^{0.1}$	Width	True 90% width
0.3	2	$[-1.30, 3.98]$	5.28	4.28
1.5	4	$[-0.15, 8.74]$	8.89	8.22
2.7	6	$[1.20, 13.48]$	12.27	12.17

Table 3: Conformal 90% prediction intervals at three representative x -values. Widths grow monotonically with x , closely tracking the oracle under the true distribution.

Intervals are slightly conservative at $x^* = 0.3$ and 1.5 , reflecting residual within-bin heterogeneity, and nearly exact at $x^* = 2.7$ (width 12.27 vs 12.17).

Empirical coverage. On a fresh test set of 2,000 observations, empirical coverage is 94.6%, 89.1%, and 79.7% at $\varepsilon = 0.05$, 0.10 , and 0.20 respectively, all within the nominal guarantees’ sampling uncertainty.

Bin boundaries and sizes. The $K^* = 6$ bins selected by cross-validation on the running example are:

$$\begin{aligned} B_1 : x \in [0.00, 0.28], \quad m_1 &= 86, & B_2 : x \in [0.28, 0.72], \quad m_2 &= 166, \\ B_3 : x \in [0.72, 1.09], \quad m_3 &= 113, & B_4 : x \in [1.09, 1.88], \quad m_4 &= 262, \\ B_5 : x \in [1.89, 2.39], \quad m_5 &= 173, & B_6 : x \in [2.40, 3.00], \quad m_6 &= 200. \end{aligned}$$

The partition adapts to the jointly growing mean and variance: the narrow bins B_1 and B_3 isolate transition regions where the mean slope and rising σ interact most strongly, while the wider bins are supported by larger sample counts.

P-value distribution. Figure 12 shows the distribution of conformal p-values $p(y^*)$ on the 2,000-point test set. The near-uniform distribution is consistent with the coverage guarantee, with slight conservatism reflecting the approximate exchangeability within bins of a heteroscedastic process.

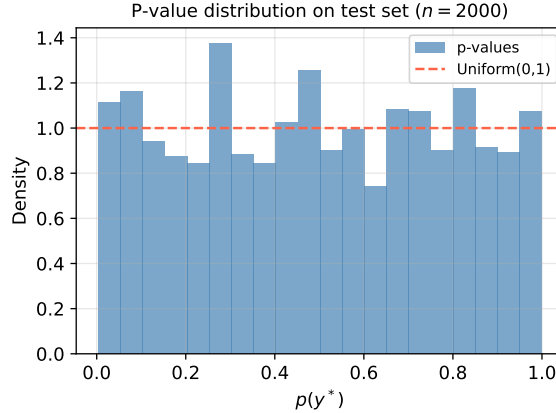


Figure 12: Distribution of conformal p-values $p(y^*)$ on the 2,000-point test set. Under a valid conformal predictor the p-values are uniform, or super-uniform (density ≤ 1 everywhere); slight conservatism reflects the approximate exchangeability within bins. P-values are discrete, taking values in $\{1/(m+1), \dots, 1\}$ for the m training points in each bin; the histogram is smoothed by the varying bin sizes across test points.

Conditional coverage. Figure 13 reports empirical coverage per bin at three levels. Bins 2–6 are close to the nominal level at all thresholds. Bin 1 (the leftmost, $m = 86$) under-covers: test points near the right edge face a true conditional distribution shifted relative to the training ECDF, an inherent price of binning a continuously varying process.

PIT histograms. Figure 14 shows the Probability Integral Transform histograms per bin. Under perfect calibration the PIT values $\hat{F}_b(y^*)$ are uniform. Bin 1 shows a U-shape (boundary effect); bins 5–6 show a mild rightward skew. Interior bins are close to

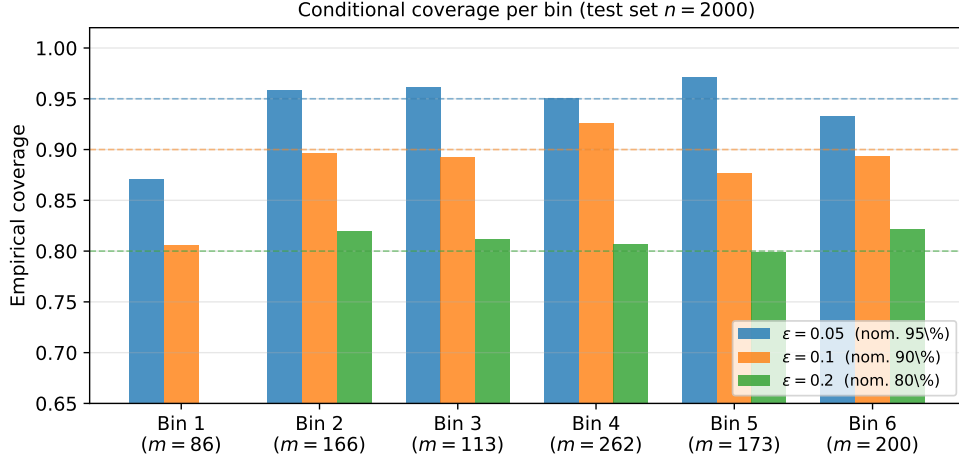


Figure 13: Conditional coverage per bin on the 2,000-point test set at three levels $\varepsilon \in \{0.05, 0.10, 0.20\}$. Dashed lines mark the nominal level. Bin 1 ($m = 86$, leftmost) under-covers due to within-bin shift of the true conditional distribution; the remaining bins are close to their nominal targets.

uniform, confirming that the CRPS-optimal partition produces well-calibrated predictive distributions where the bin size is adequate.

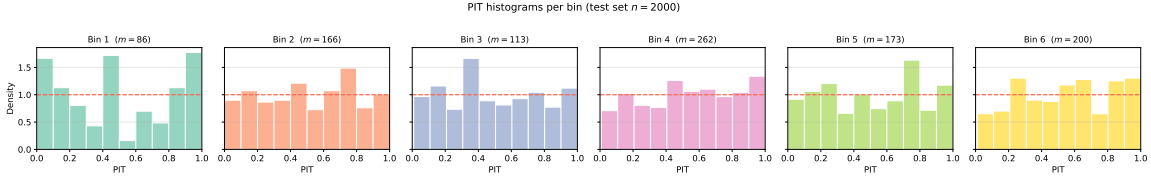


Figure 14: PIT histograms per bin. The PIT for test point (x^*, y^*) assigned to bin b is $\hat{F}_b(y^*) = m_b^{-1} \sum_{i=1}^{m_b} \mathbf{1}[y_{b,i} \leq y^*]$. Under perfect calibration the histogram is uniform (dashed line at density 1). Boundary bins (1 and 6) show departures due to within-bin heterogeneity.

Appendix G. Discussion: Scoring Rules and Cramér Distance

Other possible scoring rules. Among strictly proper rules for the full distribution, we are not aware of any other whose LOO sum reduces to a scalar computable from pairwise differences: the log score and Brier score evaluated on a fine grid would require $O(mn)$ or $O(m \log m)$ work per bin extension rather than $O(\log n)$, compromising the $O(n^2 \log n)$ precomputation. Whether the coherence between the LOO-CRPS binning criterion and the CRPS nonconformity score yields formal benefits (tighter bounds, more efficient sets relative to a mismatched pair) is an open question.

Cramér distance equivalence. A natural variant of the cross-validated criterion forms the empirical CDF $\hat{G}_n$ of all held-out observations in a bin and evaluates the predictive CDF F against it using the *Cramér distance* (Székely and Rizzo, 2013; Baringhaus and

(Franz, 2004)

$$d_C(F, \hat{G}_n) = \int_{-\infty}^{\infty} (F(t) - \hat{G}_n(t))^2 dt.$$

When $\hat{G}_n = \delta_y$ this reduces to CRPS(F, y) (Gneiting and Raftery, 2007). Expanding the square gives

$$d_C(F, \hat{G}_n) = \frac{1}{n} \sum_{i=1}^n \text{CRPS}(F, y_i) + \int_{-\infty}^{\infty} \hat{G}_n(t) [\hat{G}_n(t) - 1] dt. \quad (3)$$

The second term depends only on $\hat{G}_n$, so the correction cancels in every pairwise comparison. The two criteria are therefore identical for all purposes that matter: same minimiser, same ranking of candidate partitions, same variance of the model-selection statistic.

Appendix H. Implementation and Benchmarking Details for the CHR Comparison

We document the CHR (Sesia and Romano, 2021) implementation and benchmarking to facilitate reproducibility and to clarify the choices made in the absence of published numbers for these two datasets.

Reimplementation. No official PyPi package exists for CHR, and the reference implementation (<https://github.com/msesia/chr>) carries no LICENSE file. Rather than redistribute that code, we wrote an independent reimplementation from the published algorithm and from reading the public reference source, without copying it. The reference code implements two prediction modes on its histogram accumulator: an *interval* mode (the shortest contiguous run of histogram bins whose estimated mass reaches the target coverage) and a density-ranked *set* mode that can, in principle, return disconnected regions. It appears that only the interval mode is exercised by the reference’s own real-data experiments; consequently, we implement only the interval mode, matching what the paper actually reports.

Note that CHR requires a black-box conditional-quantile regressor. We decided to use the same `quantile_forest`-based quantile regression forest already used for CQR-QRF elsewhere in this comparison, with the identical configuration (500 trees, `min_samples_leaf`= 5) not separately tuned for CHR. This was a deliberate choice: holding the “grey-box” fixed isolates whatever the histogram/shortest-window construction contributes over plain CQR wrapping of the same regressor, rather than conflating it with a base-learner difference.

Resolution parameters. We discretize the response into 200 histogram bins (the reference’s own default, 1000, was tuned for real-dataset calibration sets of $n_{\text{cal}} = 2000$ and is unnecessary at our $n \sim 100$ –150) and use the reference’s default calibration grid resolution ($\Delta\alpha = 0.001$) for the nested-interval family used to compute conformity scores.

Randomization. The reference implementation defaults to a boundary-bin randomization step that shaves one histogram bin at the interval’s edge with a per-point random probability, sharpening coverage toward the exact nominal level. We do not use it, for consistency with the other split-conformal competitors in this comparison, none of which

use a randomized construction. It can be argued that omitting it makes split-conformal methods, CHR included, slightly conservative rather than exact.

Validation against the reference implementation. Reimplementing CHR is non-trivial, so it is critical that our implementation be validated. No published numbers exist for CHR on the two datasets used here (the original paper’s real-data experiments use a different benchmark suite), so we validated our reimplementation directly: we ran the actual reference code locally (as a correctness oracle only, never redistributed) with the identical grey-box, data, and splits. Aggregate coverage and mean width match closely and are consistent with the numbers reported above (e.g. mcycle, 15 splits: 91.4%/90.8g reference vs. 91.7%/88.0g ours). Individual intervals agree less exactly — median endpoint differences are small, but a minority of points, mostly on the smaller motorcycle calibration set, differ by tens of grams. We traced this to the shared window-search algorithm (verified identical by cross-feeding each implementation’s histogram into the other’s scoring code) being sensitive to sub-percent histogram-construction differences in sparse regions created by the grey-box’s discretized quantile predictions, amplified by the small calibration set ($n_{\text{cal}} = 67$). We believe that the resulting discrepancies do not affect coverage validity, which holds by conformal exchangeability regardless of implementation, and only marginally affect the aggregate width already reported.

Protocol. CHR is evaluated with the identical $R = 200$ random 50/50 splits, seed convention, and $\varepsilon = 0.10$ nominal level as the other split-conformal competitors in Tables 1 and 2: fit on the training half, calibrate and evaluate on the calibration half. Code is included in the accompanying repository (`experiments/chr.py`, `experiments/chr_benchmark.py`).